Experiment 8

Cursor Programming in PL/SQL
Parameterised cursors, FOR UPDATE, WHERE CURRENT OF and REF CURSOR | Oracle Database 21c XE

1. Source Code

```
-- ------------------------------------------------------------
-- Program 1: Parameterised cursor (Banking System)
-- ------------------------------------------------------------

-- 8-1a. Create the table and insert the data
DROP TABLE account PURGE;
CREATE TABLE account (
  account_no    NUMBER(6) PRIMARY KEY,
  customer_name VARCHAR2(30),
  account_type  VARCHAR2(10),
  balance       NUMBER(10,2)
);
INSERT INTO account VALUES (1001, 'Ravi', 'SAVINGS', 25000);
INSERT INTO account VALUES (1002, 'Anita', 'CURRENT', 48000);
INSERT INTO account VALUES (1003, 'Kiran', 'SAVINGS', 15000);
INSERT INTO account VALUES (1004, 'Priya', 'SAVINGS', 62000);
INSERT INTO account VALUES (1005, 'Arjun', 'CURRENT', 30000);
COMMIT;

-- 8-1b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  CURSOR c_account (p_type VARCHAR2) IS
    SELECT account_no, customer_name, account_type, balance
    FROM account
    WHERE account_type = p_type;
  v_no   account.account_no%TYPE;
  v_name account.customer_name%TYPE;
  v_type account.account_type%TYPE;
  v_bal  account.balance%TYPE;
BEGIN
  OPEN c_account('SAVINGS');
  LOOP
    FETCH c_account INTO v_no, v_name, v_type, v_bal;
    EXIT WHEN c_account%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(v_no || ' ' || v_name || ' ' ||
                         v_type || ' ' || v_bal);
  END LOOP;
  CLOSE c_account;
END;
/

-- ------------------------------------------------------------
-- Program 2: Parameterised cursor (Hospital Management)
-- ------------------------------------------------------------

-- 8-2a. Create the table and insert the data
DROP TABLE patient PURGE;
CREATE TABLE patient (
  patient_id   NUMBER(4) PRIMARY KEY,
  patient_name VARCHAR2(30),
  department   VARCHAR2(20),
  doctor_name  VARCHAR2(30)
);
INSERT INTO patient VALUES (1, 'Ravi', 'Cardiology', 'Dr. Rao');
INSERT INTO patient VALUES (2, 'Sneha', 'Neurology', 'Dr. Mehta');
INSERT INTO patient VALUES (3, 'Arjun', 'Cardiology', 'Dr. Rao');
INSERT INTO patient VALUES (4, 'Priya', 'Orthopedics', 'Dr. Singh');
INSERT INTO patient VALUES (5, 'Kiran', 'Cardiology', 'Dr. Nair');
COMMIT;

-- 8-2b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  CURSOR c_patient (p_dept VARCHAR2) IS
    SELECT patient_id, patient_name, department, doctor_name
    FROM patient
```

```
    WHERE department = p_dept;
  v_id   patient.patient_id%TYPE;
  v_name patient.patient_name%TYPE;
  v_dept patient.department%TYPE;
  v_doc  patient.doctor_name%TYPE;
BEGIN
  OPEN c_patient('Cardiology');
  LOOP
    FETCH c_patient INTO v_id, v_name, v_dept, v_doc;
    EXIT WHEN c_patient%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(v_id || ' ' || v_name || ' ' ||
                         v_dept || ' ' || v_doc);
  END LOOP;
  CLOSE c_patient;
END;
/

-- ------------------------------------------------------------
-- Program 3: FOR UPDATE cursor (Employee Payroll)
-- ------------------------------------------------------------

-- 8-3a. Create the table and insert the data
DROP TABLE employee PURGE;
CREATE TABLE employee (
  emp_id     NUMBER(4) PRIMARY KEY,
  emp_name   VARCHAR2(30),
  department VARCHAR2(20),
  salary     NUMBER(8,2)
);
INSERT INTO employee VALUES (1, 'Ravi', 'IT', 50000);
INSERT INTO employee VALUES (2, 'Anita', 'HR', 40000);
INSERT INTO employee VALUES (3, 'Kiran', 'Sales', 35000);
INSERT INTO employee VALUES (4, 'Priya', 'IT', 60000);
INSERT INTO employee VALUES (5, 'Arjun', 'HR', 45000);
COMMIT;

-- 8-3b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  CURSOR c_emp IS
    SELECT emp_id, emp_name, salary
    FROM employee
    FOR UPDATE;
  v_id     employee.emp_id%TYPE;
  v_name   employee.emp_name%TYPE;
  v_salary employee.salary%TYPE;
BEGIN
  OPEN c_emp;
  LOOP
    FETCH c_emp INTO v_id, v_name, v_salary;
    EXIT WHEN c_emp%NOTFOUND;
    UPDATE employee
    SET salary = v_salary * 1.10
    WHERE CURRENT OF c_emp;
  END LOOP;
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Salaries increased by 10% successfully.');
  CLOSE c_emp;
END;
/

SELECT * FROM employee;

-- ------------------------------------------------------------
-- Program 4: FOR UPDATE cursor (Library Management)
-- ------------------------------------------------------------

-- 8-4a. Create the table and insert the data
DROP TABLE book PURGE;
CREATE TABLE book (
  book_id          NUMBER(4) PRIMARY KEY,
  book_title       VARCHAR2(40),
  author           VARCHAR2(30),
  available_copies NUMBER(4)
);
INSERT INTO book VALUES (1, 'Database System Concepts', 'Silberschatz', 10);
INSERT INTO book VALUES (2, 'Let Us C', 'Kanetkar', 8);
INSERT INTO book VALUES (3, 'Operating Systems', 'Galvin', 6);
```

```
INSERT INTO book VALUES (4, 'Computer Networks', 'Tanenbaum', 12);
INSERT INTO book VALUES (5, 'Data Structures', 'Lipschutz', 9);
COMMIT;

-- 8-4b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  CURSOR c_book IS
    SELECT book_id, available_copies
    FROM book
    FOR UPDATE;
  v_id     book.book_id%TYPE;
  v_copies book.available_copies%TYPE;
BEGIN
  OPEN c_book;
  LOOP
    FETCH c_book INTO v_id, v_copies;
    EXIT WHEN c_book%NOTFOUND;
    UPDATE book
    SET available_copies = v_copies + 5
    WHERE CURRENT OF c_book;
  END LOOP;
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Available copies increased by 5 successfully.');
  CLOSE c_book;
END;
/

SELECT * FROM book;

-- ------------------------------------------------------------
-- Program 5: WHERE CURRENT OF (Inventory Management)
-- ------------------------------------------------------------

-- 8-5a. Create the table and insert the data
DROP TABLE product PURGE;
CREATE TABLE product (
  product_id   NUMBER(4) PRIMARY KEY,
  product_name VARCHAR2(30),
  price        NUMBER(8,2),
  quantity     NUMBER(5)
);
INSERT INTO product VALUES (1, 'Rice', 60, 100);
INSERT INTO product VALUES (2, 'Sugar', 45, 80);
INSERT INTO product VALUES (3, 'Oil', 120, 50);
INSERT INTO product VALUES (4, 'Soap', 35, 200);
INSERT INTO product VALUES (5, 'Tea', 240, 60);
COMMIT;

-- 8-5b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  CURSOR c_prod IS
    SELECT product_id, price
    FROM product
    FOR UPDATE;
  v_id    product.product_id%TYPE;
  v_price product.price%TYPE;
BEGIN
  OPEN c_prod;
  LOOP
    FETCH c_prod INTO v_id, v_price;
    EXIT WHEN c_prod%NOTFOUND;
    UPDATE product
    SET price = v_price * 1.05
    WHERE CURRENT OF c_prod;
  END LOOP;
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Product prices increased by 5% successfully.');
  CLOSE c_prod;
END;
/

SELECT * FROM product;

-- ------------------------------------------------------------
-- Program 6: REF CURSOR (University Management)
-- ------------------------------------------------------------
```

```
-- 8-6a. Create the table and insert the data
DROP TABLE student PURGE;
CREATE TABLE student (
  student_id   NUMBER(4) PRIMARY KEY,
  student_name VARCHAR2(30),
  course       VARCHAR2(20),
  marks        NUMBER(3)
);
INSERT INTO student VALUES (101, 'Rahul', 'B.Tech', 82);
INSERT INTO student VALUES (102, 'Sneha', 'B.Tech', 74);
INSERT INTO student VALUES (103, 'Arjun', 'BCA', 58);
INSERT INTO student VALUES (104, 'Priya', 'B.Sc', 91);
INSERT INTO student VALUES (105, 'Kiran', 'BCA', 45);
COMMIT;

-- 8-6b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  TYPE t_ref IS REF CURSOR;
  c_stu  t_ref;
  v_id     student.student_id%TYPE;
  v_name   student.student_name%TYPE;
  v_course student.course%TYPE;
  v_marks  student.marks%TYPE;
BEGIN
  OPEN c_stu FOR
    SELECT student_id, student_name, course, marks FROM student;
  LOOP
    FETCH c_stu INTO v_id, v_name, v_course, v_marks;
    EXIT WHEN c_stu%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(v_id || ' ' || v_name || ' ' ||
                         v_course || ' ' || v_marks);
  END LOOP;
  CLOSE c_stu;
END;
/

-- ------------------------------------------------------------
-- Program 7: REF CURSOR (Hospital Management)
-- ------------------------------------------------------------

-- 8-7a. Create the table and insert the data
DROP TABLE doctor PURGE;
CREATE TABLE doctor (
  doctor_id      NUMBER(4) PRIMARY KEY,
  doctor_name    VARCHAR2(30),
  specialization VARCHAR2(30),
  experience     NUMBER(2)
);
INSERT INTO doctor VALUES (1, 'Dr. Rao', 'Cardiology', 15);
INSERT INTO doctor VALUES (2, 'Dr. Mehta', 'Neurology', 12);
INSERT INTO doctor VALUES (3, 'Dr. Singh', 'Orthopedics', 9);
INSERT INTO doctor VALUES (4, 'Dr. Nair', 'Cardiology', 20);
INSERT INTO doctor VALUES (5, 'Dr. Iyer', 'Pediatrics', 7);
COMMIT;

-- 8-7b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  TYPE t_ref IS REF CURSOR;
  c_doc  t_ref;
  v_id   doctor.doctor_id%TYPE;
  v_name doctor.doctor_name%TYPE;
  v_spec doctor.specialization%TYPE;
  v_exp  doctor.experience%TYPE;
BEGIN
  OPEN c_doc FOR SELECT * FROM doctor;
  LOOP
    FETCH c_doc INTO v_id, v_name, v_spec, v_exp;
    EXIT WHEN c_doc%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(v_id || ' ' || v_name || ' ' ||
                         v_spec || ' ' || v_exp || ' years');
  END LOOP;
  CLOSE c_doc;
END;
/

-- ------------------------------------------------------------
```

```
-- Program 8: REF CURSOR (Online Shopping System)
-- ------------------------------------------------------------

-- 8-8a. Create the table and insert the data
DROP TABLE orders PURGE;
CREATE TABLE orders (
  order_id      NUMBER(4) PRIMARY KEY,
  customer_name VARCHAR2(30),
  product_name  VARCHAR2(30),
  quantity      NUMBER(3),
  total_amount  NUMBER(10,2)
);
INSERT INTO orders VALUES (1, 'Ravi', 'Laptop', 1, 55000);
INSERT INTO orders VALUES (2, 'Anita', 'Mobile', 2, 30000);
INSERT INTO orders VALUES (3, 'Kiran', 'Headphones', 3, 4500);
INSERT INTO orders VALUES (4, 'Priya', 'Keyboard', 1, 1500);
INSERT INTO orders VALUES (5, 'Arjun', 'Monitor', 2, 18000);
COMMIT;

-- 8-8b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  TYPE t_ref IS REF CURSOR;
  c_ord   t_ref;
  v_id    orders.order_id%TYPE;
  v_cust  orders.customer_name%TYPE;
  v_prod  orders.product_name%TYPE;
  v_qty   orders.quantity%TYPE;
  v_total orders.total_amount%TYPE;
BEGIN
  OPEN c_ord FOR SELECT * FROM orders;
  LOOP
    FETCH c_ord INTO v_id, v_cust, v_prod, v_qty, v_total;
    EXIT WHEN c_ord%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(v_id || ' ' || v_cust || ' ' || v_prod ||
                         ' ' || v_qty || ' ' || v_total);
  END LOOP;
  CLOSE c_ord;
END;
/

-- ------------------------------------------------------------
-- Program 9: Parameterised FOR UPDATE cursor (Employee Management System)
-- ------------------------------------------------------------

-- 8-9a. Create the table and insert the data
DROP TABLE employee PURGE;
CREATE TABLE employee (
  emp_id     NUMBER(4) PRIMARY KEY,
  emp_name   VARCHAR2(30),
  department VARCHAR2(20),
  salary     NUMBER(8,2),
  experience NUMBER(2)
);
INSERT INTO employee VALUES (1, 'Ravi', 'IT', 50000, 5);
INSERT INTO employee VALUES (2, 'Anita', 'HR', 40000, 3);
INSERT INTO employee VALUES (3, 'Kiran', 'IT', 45000, 4);
INSERT INTO employee VALUES (4, 'Priya', 'IT', 60000, 8);
INSERT INTO employee VALUES (5, 'Arjun', 'HR', 42000, 6);
COMMIT;

-- 8-9b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  CURSOR c_emp (p_dept VARCHAR2) IS
    SELECT emp_id, emp_name, salary
    FROM employee
    WHERE department = p_dept
    FOR UPDATE;
  v_dept VARCHAR2(20) := 'IT';
BEGIN
  FOR r IN c_emp(v_dept) LOOP
    UPDATE employee
    SET salary = r.salary + 3000
    WHERE CURRENT OF c_emp;
    DBMS_OUTPUT.PUT_LINE(r.emp_id || ' ' || r.emp_name ||
                         ' new salary: ' || (r.salary + 3000));
  END LOOP;
```

```
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Salaries updated successfully.');
END;
/

SELECT * FROM employee;

-- ------------------------------------------------------------
-- Program 10: Parameterised cursor, REF CURSOR and FOR UPDATE (College Management)
-- ------------------------------------------------------------

-- 8-10a. Create the table and insert the data
DROP TABLE student PURGE;
CREATE TABLE student (
  student_id          NUMBER(4) PRIMARY KEY,
  student_name        VARCHAR2(30),
  branch              VARCHAR2(10),
  semester            NUMBER(2),
  cgpa                NUMBER(3,1),
  scholarship_status  VARCHAR2(20)
);
INSERT INTO student VALUES (1, 'Ravi', 'CSE', 3, 9.2, NULL);
INSERT INTO student VALUES (2, 'Sneha', 'CSE', 3, 8.4, NULL);
INSERT INTO student VALUES (3, 'Arjun', 'ECE', 3, 7.8, NULL);
INSERT INTO student VALUES (4, 'Priya', 'CSE', 5, 9.5, NULL);
INSERT INTO student VALUES (5, 'Kiran', 'CSE', 5, 8.9, NULL);
COMMIT;

-- 8-10b. PL/SQL program and output
SET SERVEROUTPUT ON
DECLARE
  TYPE t_ref IS REF CURSOR;
  c_ref    t_ref;
  CURSOR c_stu (p_branch VARCHAR2) IS
    SELECT student_id, student_name, cgpa
    FROM student
    WHERE branch = p_branch
    FOR UPDATE;
  v_id     student.student_id%TYPE;
  v_name   student.student_name%TYPE;
  v_sem    student.semester%TYPE;
  v_cgpa   student.cgpa%TYPE;
  v_branch VARCHAR2(10) := 'CSE';
BEGIN
  -- display the students of the branch using a REF CURSOR
  OPEN c_ref FOR
    SELECT student_id, student_name, semester, cgpa
    FROM student WHERE branch = v_branch;
  LOOP
    FETCH c_ref INTO v_id, v_name, v_sem, v_cgpa;
    EXIT WHEN c_ref%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(v_id || ' ' || v_name || ' Sem ' ||
                         v_sem || ' CGPA ' || v_cgpa);
  END LOOP;
  CLOSE c_ref;

  -- update the scholarship status with a FOR UPDATE cursor
  FOR r IN c_stu(v_branch) LOOP
    IF r.cgpa >= 9.0 THEN
      UPDATE student
      SET scholarship_status = 'Eligible'
      WHERE CURRENT OF c_stu;
      DBMS_OUTPUT.PUT_LINE(r.student_name || ' is Eligible for scholarship.');
    END IF;
  END LOOP;
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Scholarship status updated successfully.');
END;
/

SELECT * FROM student;
```

2. Execution and Output

Output shown is sample output of the programs above (Oracle Database 21c XE, SQL*Plus).

Program 1: Parameterised cursor (Banking System)

```
[8-1a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-1b]
1001 Ravi SAVINGS 25000
1003 Kiran SAVINGS 15000
1004 Priya SAVINGS 62000

PL/SQL procedure successfully completed.
```

Program 2: Parameterised cursor (Hospital Management)

```
[8-2a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-2b]
1 Ravi Cardiology Dr. Rao
3 Arjun Cardiology Dr. Rao
5 Kiran Cardiology Dr. Nair

PL/SQL procedure successfully completed.
```

Program 3: FOR UPDATE cursor (Employee Payroll)

```
[8-3a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-3b]
Salaries increased by 10% successfully.

PL/SQL procedure successfully completed.

EMP_ID EMP_NAME DEPARTMENT SALARY
------ -------- ---------- ------
     1 Ravi     IT          55000
     2 Anita    HR          44000
     3 Kiran    Sales       38500
     4 Priya    IT          66000
     5 Arjun    HR          49500

5 rows selected.

PL/SQL procedure successfully completed.
```

Program 4: FOR UPDATE cursor (Library Management)

```
[8-4a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-4b]
Available copies increased by 5 successfully.

PL/SQL procedure successfully completed.

BOOK_ID BOOK_TITLE               AUTHOR       AVAILABLE_COPIES
------- ---------------------- ------------ ----------------
      1 Database System Concepts Silberschatz              15
      2 Let Us C                 Kanetkar                  13
      3 Operating Systems        Galvin                    11
      4 Computer Networks        Tanenbaum                 17
      5 Data Structures          Lipschutz                 14

5 rows selected.

PL/SQL procedure successfully completed.
```

Program 5: WHERE CURRENT OF (Inventory Management)

```
[8-5a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-5b]
Product prices increased by 5% successfully.

PL/SQL procedure successfully completed.

PRODUCT_ID PRODUCT_NAME PRICE QUANTITY
---------- ------------ ----- --------
         1 Rice            63      100
         2 Sugar        47.25       80
         3 Oil            126       50
         4 Soap         36.75      200
         5 Tea            252       60

5 rows selected.

PL/SQL procedure successfully completed.
```

Program 6: REF CURSOR (University Management)

```
[8-6a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-6b]
101 Rahul B.Tech 82
102 Sneha B.Tech 74
```

```
103 Arjun BCA 58
104 Priya B.Sc 91
105 Kiran BCA 45

PL/SQL procedure successfully completed.
```

Program 7: REF CURSOR (Hospital Management)

```
[8-7a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-7b]
1 Dr. Rao Cardiology 15 years
2 Dr. Mehta Neurology 12 years
3 Dr. Singh Orthopedics 9 years
4 Dr. Nair Cardiology 20 years
5 Dr. Iyer Pediatrics 7 years

PL/SQL procedure successfully completed.
```

Program 8: REF CURSOR (Online Shopping System)

```
[8-8a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-8b]
1 Ravi Laptop 1 55000
2 Anita Mobile 2 30000
3 Kiran Headphones 3 4500
4 Priya Keyboard 1 1500
5 Arjun Monitor 2 18000

PL/SQL procedure successfully completed.
```

Program 9: Parameterised FOR UPDATE cursor (Employee Management System)

```
[8-9a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-9b]
1 Ravi new salary: 53000
3 Kiran new salary: 48000
4 Priya new salary: 63000
Salaries updated successfully.

PL/SQL procedure successfully completed.

EMP_ID EMP_NAME DEPARTMENT SALARY EXPERIENCE
------ -------- ---------- ------ ----------
     1 Ravi     IT          53000          5
     2 Anita    HR          40000          3
     3 Kiran    IT          48000          4
```

```
     4 Priya    IT          63000          8
     5 Arjun    HR          42000          6

5 rows selected.

PL/SQL procedure successfully completed.
```

Program 10: Parameterised cursor, REF CURSOR and FOR UPDATE (College Management)

```
[8-10a]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[8-10b]
1 Ravi Sem 3 CGPA 9.2
2 Sneha Sem 3 CGPA 8.4
4 Priya Sem 5 CGPA 9.5
5 Kiran Sem 5 CGPA 8.9
Ravi is Eligible for scholarship.
Priya is Eligible for scholarship.
Scholarship status updated successfully.

PL/SQL procedure successfully completed.

STUDENT_ID STUDENT_NAME BRANCH SEMESTER CGPA SCHOLARSHIP_STATUS
---------- ------------ ------ -------- ---- ------------------
         1 Ravi         CSE           3  9.2 Eligible
         2 Sneha        CSE           3  8.4
         3 Arjun        ECE           3  7.8
         4 Priya        CSE           5  9.5 Eligible
         5 Kiran        CSE           5  8.9

5 rows selected.

PL/SQL procedure successfully completed.
```